



HAMK

Hämeen ammatti-
korkeakoulu



Cybersecurity

Cybersecurity--Internet-IoT Security

Ismo Turve



Internet and IoT-device security

Internet and IoT-device security

- It all depends on the user and their awareness of their own digital environment
- Environment
 - Hardware
 - Updates to information systems and equipment
 - Networks in use
- User perspective
 - Safe use of Internet
 - How to instruct family members
 - Who uses the computer

Internet and IoT-device security

- Information networks
- TCP/IP network security, TCP/IP network was not originally designed to be secure?
- Unprotected devices/systems vs. the user – The weakest link
- Internet?
 - How can I know if the other party has taken care of things?
 - Privacy – your own digital life
 - There are threats everywhere
 - Hacking is easy, for hackers
- How is security organized from a computer and network perspective?
- Unauthorized surveillance on the Internet, all unencrypted data traffic is a threat

TCP/IP Insecurity

- TCP/IP protocols were not originally designed with security in mind
- We had no idea how widespread the Internet would grow and what kind of threats there would be in the future
- Examples
 - The sender can set any source IP address to the packets they send
 - The sender of the email can set the sender's email address to anything according to the SMTP protocol
 - TCP/IP protocols don't have built-in authentication, access control, or encryption

<https://www.mailguard.com.au/blog/beware-another-fake-dropbox-phishing-scam>



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Your personal files are encrypted by CTB-Locker.



Your personal files are encrypted by CTB-Locker.

Your documents, photos, databases and other Important files have been encrypted with strongest encryption and unique key, generated for this computer.

Private decryption key is stored on a secret Internet server and nobody can decrypt your files until you pay and obtain the private key.

You only have 96 hours to submit the payment. If you do not send money within provided time, all your files will be permanently crypted and no one will be able to recover them.

Press 'View' to view the list of files that have been encrypted.

Press 'Next' for the next page.

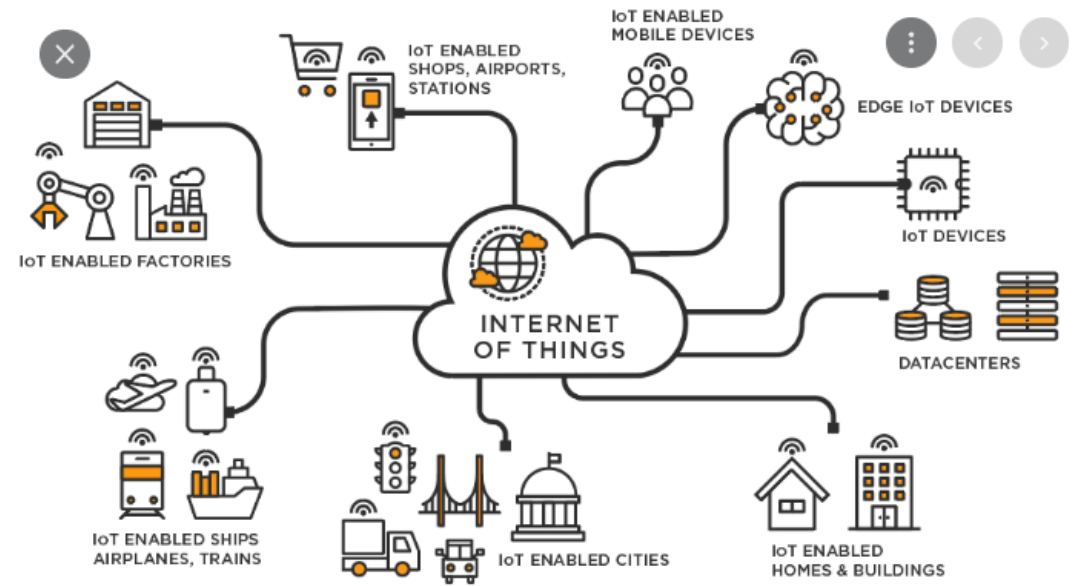


WARNING! DO NOT TRY TO GET RID OF THE PROGRAM YOURSELF. ANY ACTION TAKEN WILL RESULT IN DECRYPTION KEY BEING DESTROYED. YOU WILL LOSE YOUR FILES FOREVER. ONLY WAY TO KEEP YOUR FILES IS TO FOLLOW THE INSTRUCTION.

[View](#)**95:58:56**[Next >>](#)

IoT - Internet Of Things

- Should I be worried?
- Smart IoT devices
 - In use all the time, continuously, daily
 - Noticing vague activity?
 - Updates, do we install, do they exist?
- Passive devices are "forgotten" in the network
- Not manageable/upgradable

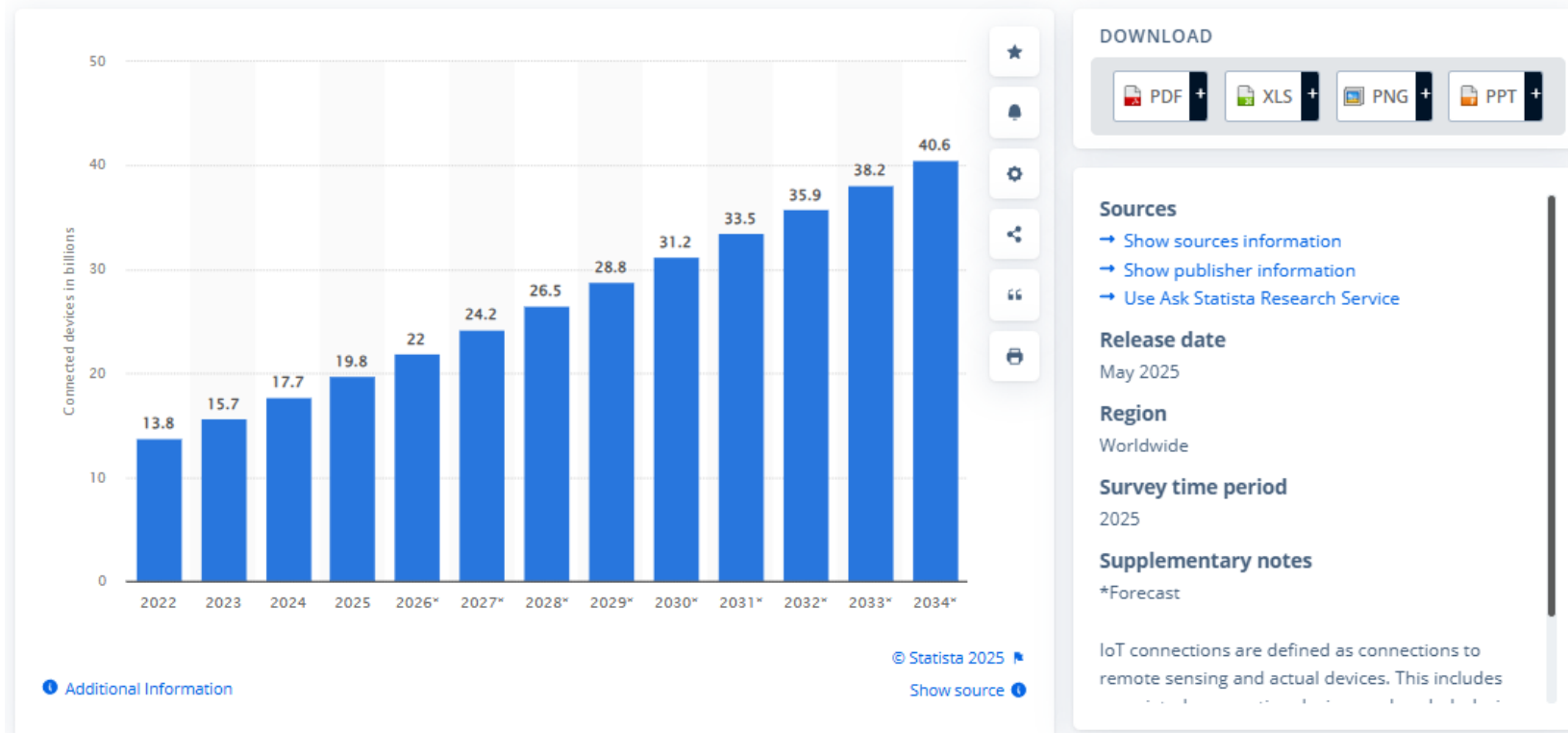


Number of Internet of Things (IoT) connected devices worldwide

<https://www.statista.com/statistics/1183457/iot-connected-devices-worldwide/>

Number of Internet of Things (IoT) connections worldwide from 2022 to 2023, with forecasts from 2024 to 2034

(in billions)



Internet of Threats

- <https://www.oxfordeconomics.com/resource/internet-of-threats/>
- <https://www.kaspersky.com/resource-center/threats/secure-iot-devices-on-your-home-network>

IoT Devices

- <https://www.softwaretestinghelp.com/iot-devices/>
- <https://cujo.com/15-most-popular-iot-products-and-devices-in-2021/>